USACE ERDC BROAD AGENCY ANNOUNCEMENT (ITL & GSL) PRE-PROPOSAL WHITE PAPER

Title: Project Ebony v8.0.0: Air-Gapped Tri-Brain OT/SCADA Safety Interlock, Automated Bridge/Pavement NDE & WORM Merkle Provenance Testbed

Submitting Organization: Humphrey Virtual Farms LLC (701 E Drew St., Madill, OK 73446)
Socioeconomic Status: Small Business, Small Disadvantaged Business (SDB), Minority-Owned, Non-Traditional Defense Contractor
SAM.gov UEI: S1M4ENLHTDH5 | CAGE Code: 1AHA8 | NAICS: 541715, 541511, 541330
Principal Investigator: Jeffery Humphrey, Founder, President & CEO | (572) 358-9978 | humphreyvirtualfarm@gmail.com
ERDC Topic Areas: ITL-4 (Cybersecurity), ITL-8 (Edge Computing), GSL-19 (Bridge Research), GSL-21 (Military Structures), GSL-23 (Advanced Maneuver / Airfield Pavements)
Research Classification: Applied Research and Development (TRL 4 Laboratory Software Testbed advancing to TRL 6 Prototype; Non-Proprietary Pre-Proposal Summary)
Cross-Referenced DoD CDAO Tradewinds Docket: 10-26-4046 (Status: Received | Video SHA-256: c597865785eae6fdad6fdf6803a2710aaa0fb5c2fa28fe512b6ed76ef8b7a7fd | Merkle Root: bb39eeaa66a39ad2e0c0ab74ba4a5c4faac2bf0d98babc0362482b97607151a1)
Public Verification Repository: https://github.com/mrshumphrey3251-ai/hvf-media-matrix-public

1. Problem Identification, Scientific Scope & USACE ERDC Mission Impact

Per U.S. Government Accountability Office audits GAO-21-81 and GAO-24-106137, the Department of Defense and U.S. Army Corps of Engineers (USACE) operate over 500 military installations, 300,000+ buildings, civil works locks and dams, and $4.2B+ in operational technology (OT) energy infrastructure relying on commercial Programmable Logic Controllers (PLCs) scanning at 50 to 200 milliseconds. Per Joint CISA/NSA/FBI/DoD Advisories AA23-144A and AA24-038A (Volt Typhoon) and TRISIS/Triton analyses, state-sponsored actors pre-position inside critical infrastructure OT networks to corrupt PLC safety logic while spoofing operator displays. Because high-voltage arc-flash backfeeds destroy switchgear in under 4.16 milliseconds and thermal excursions occur in 6.0 milliseconds, legacy 50 to 200 millisecond PLC scans react 8x to 33x too late. Simultaneously, conventional civil and military bridge and airfield pavement inspections rely on manual lane closures and snooper trucks costing $4,600+ per day without cryptographic chain-of-custody verification.

2. Technical Approach: Air-Gapped Tri-Brain Architecture (TRL 4 Baseline to TRL 6)

Project Ebony v8.0.0 is an active TRL 4 laboratory software emulation testbed operating 100% offline on local workstation loopback 127.0.0.1 with 0% commercial cloud dependency, 0 external retrieval APIs, 0 third-party vector databases, and 0 open WAN ports:

Brain 1 - Deterministic OT/SCADA Safety Interlock (ITL-4 / ITL-8): Executes deterministic physical boundary mathematics in software and models our Phase-1 hardware target: a sub-50 microsecond bare-metal Cortex-M7/RISC-V control loop, a 2.04 microsecond RS-485 Modbus FC05 trip, and a 36.75 nanosecond Silicon-Carbide (SiC) MOSFET Analog Kinetic Guillotine gate cutoff (1,360,544x faster than a 50ms PLC at a 28.5W target power draw on a 24V DC bus).

Brain 2 - Isolated Bridge & Airfield Pavement NDE Diagnostics (GSL-19 / GSL-21 / GSL-23): Operates across a unidirectional read-only buffer with zero electrical actuator wire. Performs offline automated ASTM D6433 Pavement Condition Index (PCI) distress classification, crack-width mensuration, and civil/military bridge structural health monitoring, modeling a 94.6% inspection cost reduction ($250/sortie autonomous drone NDE vs. $4,600/day manual snooper-truck closures).

Brain 3 - Zero-Dependency 52-Table SQLite WORM Enclave & Merkle Provenance (ITL-4 / ITL-8): Executes all telemetry and anomaly retrieval locally inside an air-gapped 52-table bare-metal SQLite Write-Once-Read-Many (WORM) enclave (matrix_ledger.db and hvf_memory_vault.db) and chains every frame into a SHA-256 Merkle DAG in under 0.15ms across 637 compiled C2 Python modules, 85/85 passing unit test suites, 17/17 blocked Red-Team cyber vectors with automatic single-byte tamper zeroization, and a completed 110-control NIST SP 800-171 Rev 2 and 800-82 Rev 2 mapping.

3. Quantitative Analysis of Alternatives (Advancing the State of the Art)

Across six mission gates (sub-millisecond physical cutoff, zero-cloud air-gap sovereignty, cryptographic WORM provenance, hardware-isolated NDE diagnostics, <30W SWaP-C budget, and automated tamper zeroization), Project Ebony v8.0.0 passes 6 of 6 gates. Legacy PLCs (Siemens S7-1500, Rockwell ControlLogix) require 50-200ms and draw 150-450W; Schneider Triconex SIS requires 10-50ms, draws 200-600W, and was compromised by TRISIS firmware injection; passive IDS platforms (Dragos, Nozomi) draw 300-1,200W with zero physical cutoff authority; and commercial cloud AI wrappers require >250ms round-trip latency and fail under EMCON or SATCOM severance.

4. Proposed 12-Month Applied Research & Development Plan & Rough Order of

Magnitude
Humphrey Virtual Farms LLC proposes a 12-Month, $1,200,000.00 Fixed-Price Applied Research and Development Effort (0% Cost Share) advancing Project Ebony from TRL 4 to TRL 6:
- Milestone 1 (Months 1-2 | $150,000.00): ERDC ITL/GSL Interface Control Document (ICD) & 52-Table WORM Enclave Baseline.
- Milestone 2 (Months 3-6 | $350,000.00): Brain 1 Bare-Metal Cortex-M7 + SiC MOSFET Prototype Fabrication & <50us / 36.75ns Oscilloscope Verification.
- Milestone 3 (Months 7-9 | $350,000.00): Brain 2 Offline ASTM D6433 Airfield Pavement & Bridge NDE Calibration with ERDC GSL Datasets.
- Milestone 4 (Months 10-12 | $350,000.00): TRL 6 Integrated GSL/ITL Hardware-in-the-Loop Demonstration, C3PAO/SPRS Package & Final Technical Report.
Compliance & Rights Statement: This pre-proposal contains no classified or restricted proprietary data. Background software testbed modules developed at private expense are identified under DFARS 252.227-7018 / FAR 52.227-14.